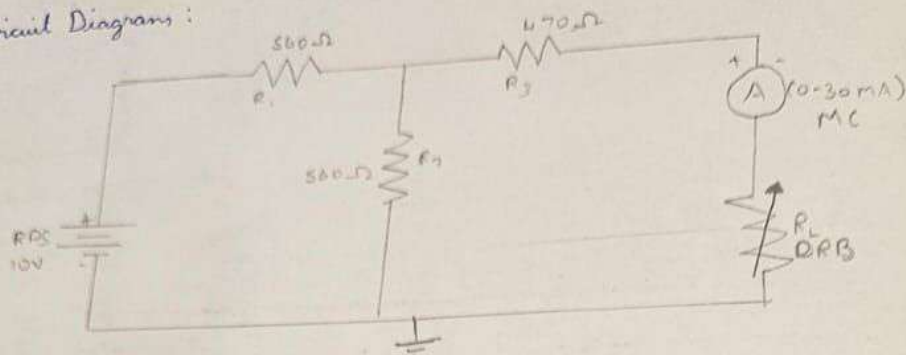
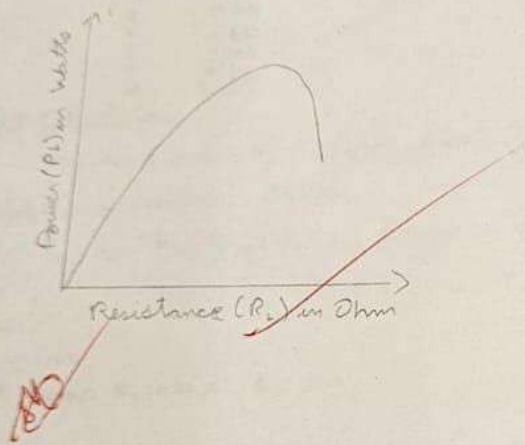


Circuit Diagram:



Model Graph:



Aim: To measure the value of load resistor.

Apparatus used:

S. No	Name
01	
02	
03	
04	R

Procedure:

1. Make Connections.
2. Change the load resistor and note the corresponding current.
3. Plot a graph of Power (P) vs Resistance (RL).
4. Verify the result.

Resistance (RL) in Ohm	Current (I) in mA	Power (P) in mW	Remarks
10	0.0	0.0	Initial reading
20	0.0	0.0	Initial reading
30	0.0	0.0	Initial reading
40	0.0	0.0	Initial reading
50	0.0	0.0	Initial reading
60	0.0	0.0	Initial reading
70	0.0	0.0	Initial reading
80	0.0	0.0	Initial reading
90	0.0	0.0	Initial reading
100	0.0	0.0	Initial reading

Result: The value of load resistor is 50Ω.

Exp. no: 06

Date: 24/02/26

Verification of maximum power transfer theorem

Aim: To measure the power absorbed in a load and to verify that the absorbed power in a load is maximum only when load resistance is equal to the source resistance.

Apparatus required:

S.NO	Name of the apparatus	Range / Rating	Quantity
01	Voltmeter	(0-15V) MC	1
02	Ammeter	(0-500mA) MC	1
03	Resistors	560 Ω , 470 Ω	2
04	RPS (DC supply)	15V	1

Procedure:

1. Make connections as per the diagram
2. Change the Resistor R_L , whose value close to R_{TH} , measure the corresponding value V_L
3. Plot a graph between R_L and P_L and find the R_L
4. Verify the measured values of P_L at maximum power transfer

Table (1): For Theoretical Calculation: $R_{th} = 750 \Omega$

S.no	Load Resistance (R_L) in ohm	Load Current (I_L) in mA	Load Voltage (V_L) in volts	Load Power (P_L) in watts
				0.0020
01	500	4.00	2.00	0.00335
02	750	3.33	2.50	0.00316
03	1000	2.80	2.80	0.00606
04	1500	2.00	3.00	0.00336
05	2000	1.67	3.33	

Table (2): for practical Calculation

S.no	Load Resistance (R_L) in ohm	Load Current (I_L) in Amps	Load Voltage (V_L) in volts	Load power (P_L) in watts
				0.0076
01	500	3.90	1.90	0.00793
02	750	3.25	2.44	0.00784
03	1000	2.80	2.30	0.00571
04	1500	1.95	2.93	0.00512
05	2000	1.66	3.20	

Model Calculation:

Given,

$$R_1 = 560 \Omega$$

$$R_2 = 560 \Omega$$

$$R_3 = 470 \Omega$$

$$\text{Supply Voltage} = 20V$$

Step 1: Find R_{TH}

$$R_{TH} = R_3 + \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$

$$= 470 + \frac{560 \times 560}{560 + 560}$$

$$= 470 + 280$$

$$R_{TH} = 750 \Omega$$

Step 2:

Step 3:

$$R_L = R_{TH}$$

$$R_L = 750 \Omega$$

S.no	
01	
02	
03	
04	
05	

Result
Th
and

Step 2: find V_{TH}

$$V_{TH} = 10 \times \frac{R_2}{R_1 + R_2}$$

$$= 10 \times \frac{560}{560 + 560}$$

$$= 5V$$

Step 3: $R_L = 750 \Omega$

$$R_L = R_{TH}$$

$$R_L = 750 \Omega$$

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

$$= \frac{5}{750 + 750}$$

$$= 3.33 \text{ mA}$$

$$= 0.00333 \text{ A}$$

$$V_L = I_L \times R_L = 0.0033 \times 750$$

$$= 2.5$$

$$P_L = V_L \times I_L$$

$$= 2.5 \times 0.0033$$

$$= 8.44 \text{ mW}$$

S.no	Description	Marks Allotted	Marks obtained
01	Experimental Setup	20	20
02	Observation / Reading	20	20
03	Calculation / Result	20	20
04	Viva Qs	20	20
05	Record	20	20
	Total	100	100

100

24/12/20

Result:

Thus maximum power transfer theorem is verified practically and theoretically